Rashad Hamidi

PhD Researcher in Theoretical Physics

xurashad@gmail.com • xurashad.github.io • 0000-0002-0719-3269 • InspireHEP

Unite Students – Newgate Court, Grainger St, Newcastle upon Tyne, UK, NE1 5AT

Profile & Current Status

Current Status: PhD student in Mathematical and Theoretical Physics group at Durham University, UK.
Supervisor: Prof. Ben Hoare **Expected Completion:** August 2026

Research Interests: Integrable Models, String Theory, Quantum Field Theory.

My primary research interests lie in the areas of integrable systems, string theory and quantum field theory. I am particularly focused on exploring integrable sigma models, their deformations, and their connections to higher-dimensional gauge theories, such as 4d Chern-Simons theory. My work involves studying the mathematical structures underlying these models, including quantum groups and Yang-Baxter equations, as well as investigating their physical applications.

Study Experience

PhD in Mathematical and Theoretical Physics [2022 – Expected 2026]
Department of Mathematical Sciences, Durham University, UK

MSc in Particles, Strings and Cosmology [2021 – 2022]
Department of Mathematical Sciences, Durham University, UK

BSc in Physics (Minor in Mathematics) [2017 – 2021]
Department of Physics, Birzeit University, Palestine

Research Papers & Theses

- **Hamidi, R.**, Hoare, B. *Twists of trigonometric sigma models.* High Energ. Phys. **2025**, 90 (2025). doi:10.1007/JHEP08(2025)090
- **Hamidi, R.** *BTZ Black Hole: Holographic Duality, Entropy, and AdS_3/CFT_2.* MSc Thesis supervised by Iqbal, N. 2022.
- **Hamidi, R.** *Fractional Quantum Mechanics.* BSc Thesis supervised by Sayyed-Ahmad, A. 2022.

Talks and Posters

- **IDD25:** Poster presentation at *Integrability, Dualities and Deformation (2025 Edition)*, Nordita, Stockholm, 2025: "Integrable 2d trigonometric $\mathbb{Z}_N$-twisted σ-models."
- **NBMPS74:** Talk at *North British Mathematical Physics Seminars (74th Edition)*, York University, 2025: "Twists of trigonometric sigma models."
- **HEP4P:** Online talk at *High-Energy Physics for Palestine Seminars*, 2024: "Integrable models form 4d Chern-Simons action."
- **BZU PHYS Seminars:** Talk at *Department of Physics Seminars*, Birzeit University, 2021: "Fractional Quantum Mechanics."

Work Experience

- **Teaching Assistant** [2023 – Present]
  Department of Mathematical Sciences, Durham University
  - *Epiphany 2026:* Discrete Mathematics.
  - *Epiphany 2026:* Maths for Engineers and Scientists.
  - *Epiphany 2026:* Analysis I.
  - *Michaelmas 2025:* Discrete Mathematics.
  - *Epiphany 2025:* Single Math A, Single Math B.
  - *Michaelmas 2024:* Single Math A, Single Math B.
  - *Epiphany 2024:* Mathematical Physics II.
  - *Epiphany 2024:* Special Relativity & Electromagnetism II.
  - *Michaelmas 2023:* Mathematical Physics II.
  - *Epiphany 2023:* Special Relativity & Electromagnetism II.

- **Astronomical Observatory Tour Guide** [2017 – 2021]
  Michel and Sanieh Hakim Observatory, Birzeit University
  - Led engaging and educational guided tours of the observatory facilities for diverse audiences.
  - Developed and delivered accessible presentations on astronomical concepts and celestial objects.
  - Operated the main telescope to showcase planets, stars, nebulae, and galaxies during public nights.
  - Assisted in planning, promotion, and execution of public outreach events (eclipse viewings, meteor showers).

Groups and Memberships

- Ben Hoare Integrability Group, Durham University [2022 – Present]
- Mathematical and Theoretical Physics Group, Durham University [2022 – Present]
- High Energy Physics for Palestine Group [2022 – Present]
- Center of Particle Theory (CPT), Durham University [2021 – Present]
- Physics Club & Astronomy Club, Birzeit University [2017 – 2021]

Honors, Awards & Scholarships

- STFC Scholarship for PhD research, Durham University (2022)
- Distinction certificate for MSc, Durham University (2022)
- Global Masters Scholarship, Durham University (2021)
- Distinction certificate for BSc, Birzeit University (2021)
- Mosa Naser Scholarship for Physics, Birzeit University (3 years: 2018/2019, 2019/2020, 2020/2021)
- Science Faculty Scholarship, Birzeit (2017/2018)
- Paltel Scholarship for undergraduate studies (2017)
- Honor Scholarship, Birzeit University (7 terms: Spring 2018, Fall 2018, Spring 2019, Fall 2019, Spring 2020, Fall 2020, Spring 2021)
- Honor list for 8 consecutive terms at Birzeit University (Fall 2017 – Spring 2021)

Skills

- **Scientific Computing & Mathematics:** Mathematica, Python, C, MATLAB, Octave, R, LaTeX, TikZ, Numerical Methods, Tensor Calculus.
- **Web & Software Development:** React, Next.js, TypeScript, Node.js, HTML/CSS, Tailwind CSS, PHP, Bash Scripting, API Integration, Git.
- **AI & Data Science:** Generative AI, LLMs, Prompt Engineering, Dataset Curation, Data Visualization.
- **Modelling & Design:** Unity, Blender, Unreal Engine, AutoCAD (2D/3D).
- **Electronics & Astronomy:** Arduino prototyping, LabVIEW, RSpec, Stellarium, Celestia.
- **Languages:** Arabic (Native), English (Fluent).

Conferences, Workshops and Schools

- Integrability in Quantum Field Theory, Durham University, 2026.
- North British Mathematical Physics Seminars (6 times: 75 Durham 2025, 74 York 2025, 72 Durham 2024, 71 York 2024, 69 Durham 2023, 66 Durham 2022).
- Integrability, Dualities and Deformations Conference (3 times: Nordita 2025; Swansea 2024; Durham 2023).
- Integrability in Gauge and String Theory (3 times: DESY Hamburg 2026, King's College London 2025, ETH Zurich 2023).
- Durham Hackathon (DurHack) 2024, Durham University, 2024.
- Paths to Quantum Field Theory Workshop (2 times: Durham 2023, Durham 2022).
- Young Researchers Integrability School and Workshop, Durham University, 2023.
- Meteorology Conference, Birzeit University, 2020.
- Palestinian Advanced Physics School by Scientists for Palestine, Birzeit University, 2019.
- Condensed Matter Conference, Birzeit University, 2019.
- Hands-on Molecular Dynamics Workshop by Palestinian-German Science Bridge, Birzeit University, 2019.
- Computational Multiphysics Conference, Birzeit University, 2019.
- Physics Without Frontiers Workshop by Abdus Salam International Centre for Theoretical Physics (ICTP), Birzeit University, 2017.
- Black Holes Conference by Harvard-Smithsonian Center for Astrophysics, Birzeit University, 2017.

Pre-University Participation

- Access Programme by AMIDEAST, 2014-2016.
- Physics Olympiad, Ramallah, 2015.
- Science Exhibition, Tulkarm, 2015.
- Science and Technology Entrepreneurship Program (STEP), Ramallah, 2014.
- First Lego League (FLL) Robotics, 2013 & 2014.
- Intel ISEF, Ramallah, 2012.
- SyscoLab Competition, Ramallah, 2012.
- SEEK Electronic Bag Training, Ramallah, 2012.

Developed Applications

You can access my developed applications through my website: https://xurashad.github.io/apps

Scientific Applications

- **Particle Physics Visualizer:** A web app provides an interactive supersymmetric particles periodic table.
- **Interactive Periodic Table:** A web app provides an interactive periodic table with detailed information about each element.
- **Fractal Explorer:** A tool for generating the Mandelbrot and Julia sets with customizations.
- **Feynman Diagram Generator:** A tool for creating Feynman diagrams easily and intuitively, and exporting the diagrams as TikZ LaTeX code.
- **Graphing Calculator:** A tool for creating and visualizing mathematical graphs easily and intuitively.
- **Celestial Simulator:** A tool for simulating the solar system and visualizing the orbits of planets, moons, and other celestial bodies. It includes features such as time control, zooming, and information display about each object.
- **Earth Explorer:** An interactive 3D globe powered by CesiumJS and OpenStreetMap.

Datasets Generators

- **Fibonacci Sequence Generator:** A tool for generating and downloading the Fibonacci sequence in JSON format. For fast generation, it is recommended to generate the sequence up to the first 50,000 numbers, or the nth number up to 1,000,000. Limits can be unlocked.
- **Prime Sequence Generator:** A tool for generating and downloading the prime sequence in JSON format. For fast generation, it is recommended to generate the sequence up to the first 1,000,000 numbers. Limits can be unlocked.

Professional Applications

- **Website Builder Pro:** A user-friendly professional app to build up websites without coding.
- **CV Builder Pro:** A user-friendly professional app to build up CVs without coding.
- **LaTeX Editor Pro:** A user-friendly professional app to write LaTeX documents without coding.
- **Office Suite Pro:** A user-friendly professional app that includes a word processor, a spreadsheet editor, and a presentation maker. It is designed to be a free and open-source alternative to Microsoft Office and Google Docs. It supports text formatting, tables, charts, image insertion, video streaming, PDF presentation, HTML page screening, python code execution, bash scripting, and more.

AI Applications

- **AI Chatbot:** A multimodal LLM-based chatbot that can understand and generate text.
- **ArtForge AI Image Generator:** A multi-style AI image generator for AI-assisted art creation.

Utilities

- **Plagiarism Checker:** A tool for checking the originality of the text and detecting plagiarism.
- **Unicode Explorer:** A tool for searching the library of unicode characters.
- **Unicode Font Generator:** A tool for generating text in various Unicode font styles.
- **PDF Tools:** An all-in-one tool for PDF. It can be used for extracting pages from the PDF, convert image to PDF or PDF to image, and merge PDF documents.
- **Multimedia Base64 Converter:** A tool for encoding files to base64 and decoding base64 strings back to files.

- **QR Code Generator:** A tool for generating QR codes easily and quickly with customizations.
- **Image Editor:** Professional-grade image editor with crop, resize, rotate, flip, 12 filters, 10 adjustment sliders, freehand drawing, text overlays, and multi-format export.

Games

- **2048 Tile:** A game that requires sliding numbered tiles on the grid to combine them and create a tile with the number 2048.
- **Celestial Chess:** A strategic chess game set in a celestial theme.

Developed Datesets

You can access my developed datasets through my website: https://xurashad.github.io/datasets

- **Fundamental & Supersymmetric Particles:** A dataset containing information about fundamental particles and their supersymmetric counterparts. Covers the Standard Model particle zoo alongside their SUSY partners — squarks, sleptons, gauginos, and more.
- **First 1M Prime Numbers:** A dataset of the first 1,000,000 prime numbers, from 2 to 15,485,863. Generated via an optimised Sieve of Eratosthenes. Useful for number theory research, cryptography benchmarks, and machine learning experiments.
- **Chemical Elements:** A comprehensive dataset of all chemical elements in the periodic table. Each entry includes physical properties, thermodynamic constants, electron configuration, and a Bohr model image.

Course-Based Projects

- BTZ Black Hole: Holographic Duality, Entropy, and AdS_3/CFT_2, *Particles, Strings and Cosmology*, summer 2022.
- Fractional Quantum Mechanics, *Quantum Mechanics*, spring 2021.
- Computational Planetary Motion, *Computational Physics*, fall 2020.
- Many Body Problem, *Computational Physics*, fall 2020.
- Cellular Automata, *Computational Physics*, fall 2020.
- Interactive Visualization of the Logistic Map, *Computational Physics*, fall 2020.
- Stellar Spectroscopy – Rigil, Betelgeuse and Sirius, *Astronomy*, spring 2020.
- Magnetic Fluids, *Nanophysics*, fall 2019.
- Vantablack - Optical Coating and 3D Light Microscopy, *Medical Physics*, spring 2019.

Selected Coursework

- **MSc in Particles, Strings and Cosmology (Durham University)**
  - **Quantum Field Theory:** Introductory Field Theory, QFT I & II, QED, Supersymmetry.
  - **General Relativity & Cosmology:** General Relativity, Cosmology I & II, Neutrino Physics, Astroparticle Physics.
  - **Gauge Field Theory:** Group Theory, Standard Model, Renormalisation Group, Scattering Amplitudes, Non-Perturbative Physics.
  - **Superstring Theory:** Conformal Field Theory, String Theory.
  - **Phenomenology:** QCD, Flavour Physics and Effective Field Theories, Higgs Physics.
  - **Extra:** Introduction to AdS/CFT, Analytic loop integration.
- **BSc in Major Physics Minor Mathematics (Birzeit University)**
  - **Physics:** General Physics I–III, Modern Physics, Medical Physics, Astronomy, Vibrations & Waves, Optics, Analog Electronics, Mathematical Physics, Classical Mechanics I & II & master, Electromagnetic Theory I & II, Quantum Mechanics I & II, Thermal & Statistical Mechanics, Computational Physics.
  - **Physics Labs:** General Physics I–III Lab, Modern Physics Lab, Analogue Electronics Lab, Advanced Physics Lab.
  - **Mathematics:** Calculus I–III, Foundations of Mathematics (Logic), Linear Algebra I & II, ODEs, PDEs, Numerical Methods, Abstract Algebra I, Mathematical Analysis I.
  - **Statistics & Computer Science:** Statistics I & III (Probability Theory), Programming C.
  - **Extra Scientific Courses:** General Chemistry I, General Biology I.
  - **Extra Scientific Labs:** General Chemistry Lab I, General Biology Lab I.

– **Extra Philosophy Courses:** Modern and Contemporary European Civilisation, Modern and Contemporary Arab Thought.

Referees

Name	Affiliation	Email	Webpage
Prof. Ben Hoare	Durham University	ben.hoare@durham.ac.uk	www.benhoare.info
Dr. Ana Retore	DESY	ana.retore@desy.de	
Prof. Nabil Iqbal	Durham University	nabil.iqbal@durham.ac.uk	www.nabiliqbal.com
Prof. Benoît Vicedo	York University	b.vicedo@york.ac.uk	
Prof. Hazem Abusara	Birzeit University	habusara@birzeit.edu	